List of topics for Midterm:

1 Constant coefficients transport equations

Find general solution.

Both Linear

- homogeneous
  - with no lower order terms
    $$a u_x + b u_y = 0$$
  - with lower order terms
    $$a u_x + b u_y + c u = 0$$
    constant
- Inhomogeneous problem $\longrightarrow$ $a u_x + b u_y = d$ (easier).
  - $a u_x + b u_y = F(x,y)$ (Duhamel integral)

2 To all the problems in 1 we can impose initial conditions to one solution among many.

Remark: $a u_x + b u_y + d u = f(x,y)$ (*)

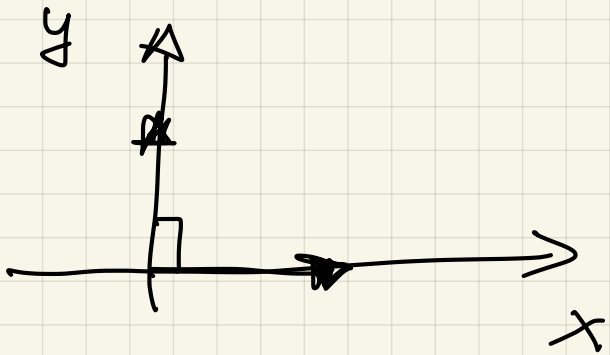
$\longrightarrow$ $\boxed{a u_x + b u_y + d u = 0}$ homogeneous eq.

$\hookrightarrow$ homog. sol.

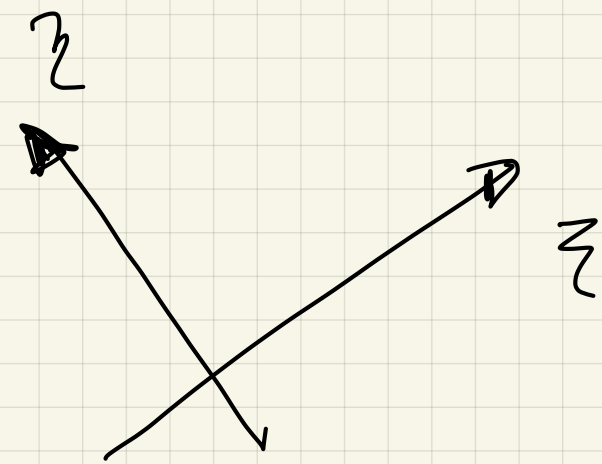
Duhamel formula uses $f(x,y)$ to find the special solution.

In the end the sol (*) = sol. homog + special sol.

Coordinate method ideas



change variable $\longrightarrow$



$$a u_x + b u_y$$

$$\vec{v} = (a, b)$$

$$\begin{cases} \xi = ax + by = \vec{v} \cdot (x, y) \\ \eta = bx - ay = \vec{v}^{\perp} \cdot (x, y) \end{cases}$$

3 Variable coefficient transport (like practice problem 6)

$$\sqrt{1-x^2}\, u_x + u_y = 0 \qquad u(0, y) = y$$

$$u_x + \frac{1}{\sqrt{1-x^2}}\, u_y = 0 \qquad \text{if } \boxed{|x| < 1}$$

correct condition

$|x| \leq 1$

$\longrightarrow$ if $x = 1$ then $u_y = 0 \rightarrow u(x, y) = f(x) + C$ impossible

Then $\dfrac{dy}{dx} = \dfrac{1}{1-x^2} \longrightarrow \dfrac{du}{dx} = 0$

$y(\xi, 0) = \xi$

$$y = \arcsin x + \xi$$

$$\xi = y - \arcsin x$$

$$\boxed{u(x,y) = f(y - \arcsin x)}$$

$u(0,y) = y \qquad f(y) = y$

$$\boxed{u(x,y) = y - \arcsin x}$$

$u_x = -\dfrac{1}{\sqrt{1-x^2}} \qquad u_y = 1$

4 Inhomogeneous Transport: Duhamel formula

something like practice problem 5

5 Wave Eq. $\longrightarrow$ something like practice problem 8

6 Wave Eq. $\rightarrow$ D'Alambert formula, something like practice problem 9

7 something on energy for wave or damped wave or wave attached to coil (as in class). But <u>not</u> uniqueness yet (<u>nothing</u> like 10b) yet)

- YOU SHOULD KNOW HOW TO integrate by parts in higher dimensions

$$u_{tt} - c^2 \Delta u = 0 \qquad \frac{d}{dt}(\nabla u \cdot \nabla u) = \underbrace{2 \nabla_x u \cdot \nabla_x u_t}$$

$$E[u](t) = \frac{1}{2} \int_{\mathbb{R}^d} |u_t|^2 + |\nabla_x u|^2 \, dx$$

$$\frac{dE}{dt} = 0 \qquad \underbrace{E[u](t)}_{\text{constant in time.}} = E[u](0)$$

$$\operatorname{div} \nabla = \Delta$$

$$\int |u_t|^2 + |u_x|^2 \, dx$$

You had to do this in Additional Problem 3, Hmwk set 6.

WON'T ASK YOU ABOUT THIS YET

Uniqueness using energy conservation or decreasing:

$$0 \leq E(u)(t) \leq E(u)(0)$$

$\hookrightarrow t=0$

$$\frac{1}{2}\int \underbrace{|u_t(x,0)|^2}_{} + \underbrace{|u_x(x,0)|^2}_{} dx$$

$$\frac{1}{2}\int |\psi(x)|^2 + |\phi'(x)|^2 dx$$

Assume u_1 and u_2 two solutions for <u>same initial data</u>. Call $w = u_1 - u_2$. If equation is linear $\Rightarrow$ w is also a solution with initial data 0.

$$\begin{cases} \Box u_1 = 0 \\ u_1(x,0) = \phi \\ \partial_t u_1(x,0) = \psi \end{cases} \qquad \begin{cases} \Box u_2 = 0 \\ u_2(x,0) = \phi \\ \partial_t u_2(x,0) = \psi \end{cases}$$

$$w = u_1 - u_2 \qquad \begin{cases} \Box w = 0 \\ \boxed{\begin{aligned} w(x,0) &= 0 \\ w_t(x,0) &= 0 \end{aligned}} \end{cases} \quad (*) \qquad E[w](0) = 0$$

$$0 \le E[w](t) \overset{\le}{=} E[w](0) = 0$$

$$\Downarrow \forall t \qquad \forall t$$

$$\frac{1}{2}\int_{\mathbb{R}} \underbrace{|w_t(x,t)|^2}_{=0} + \underbrace{|w_x(x,t)|^2}_{=0}\, dx \implies \underline{w(x,t) = 0}$$

use $(*)$